Structural Break and Time Series Modeling of Weekly Cereal Sales

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Introduction

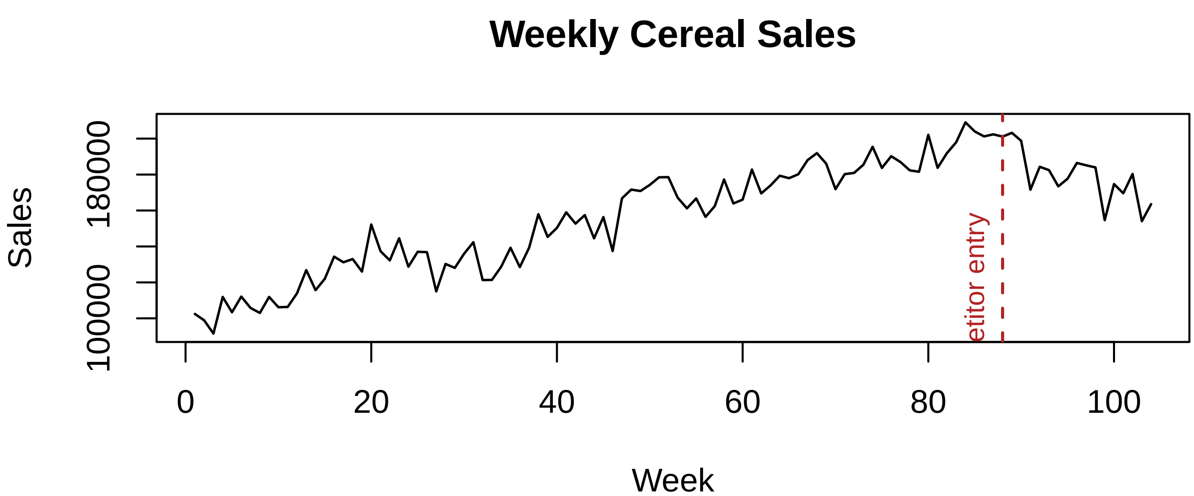
This report examines a published example containing 104 weeks of cereal sales. In the scenario accompanying the data, a competing product enters the market in week 88. The analysis asks whether the sales level or weekly trend changed at that point while accounting for serial correlation.

The dataset comes from the weekly cereal-sales intervention example in Montgomery, Jennings, and Kulahci's *Introduction to Time Series Analysis and Forecasting*, Second Edition, and is available through Wiley's companion materials. The source does not identify the brands or establish whether the values are observed or simulated.

The analysis has three goals:

1. Estimate the immediate level change and the change in trend at week 88.
2. Model autocorrelation rather than relying on ordinary least-squares inference.
3. Produce short-term model-based forecasts and describe their limitations.

Data and Exploratory Analysis



Sales rise through most of the series and then decline after week 88. A segmented trend is therefore a natural description of the visible change. The event timing is supplied with the example, so the analysis treats week 88 as known rather than searching the series for a break point.

The design is an interrupted time series without a control series or additional covariates. It estimates a change in the brand's weekly sales associated with the stated event date, but it cannot establish that competitor entry caused the change; market share is not observed.

Segmented Regression

Let I_t equal one beginning in week 88, and let $A_t = \max(0, t - 88)$. The deterministic part of the model is

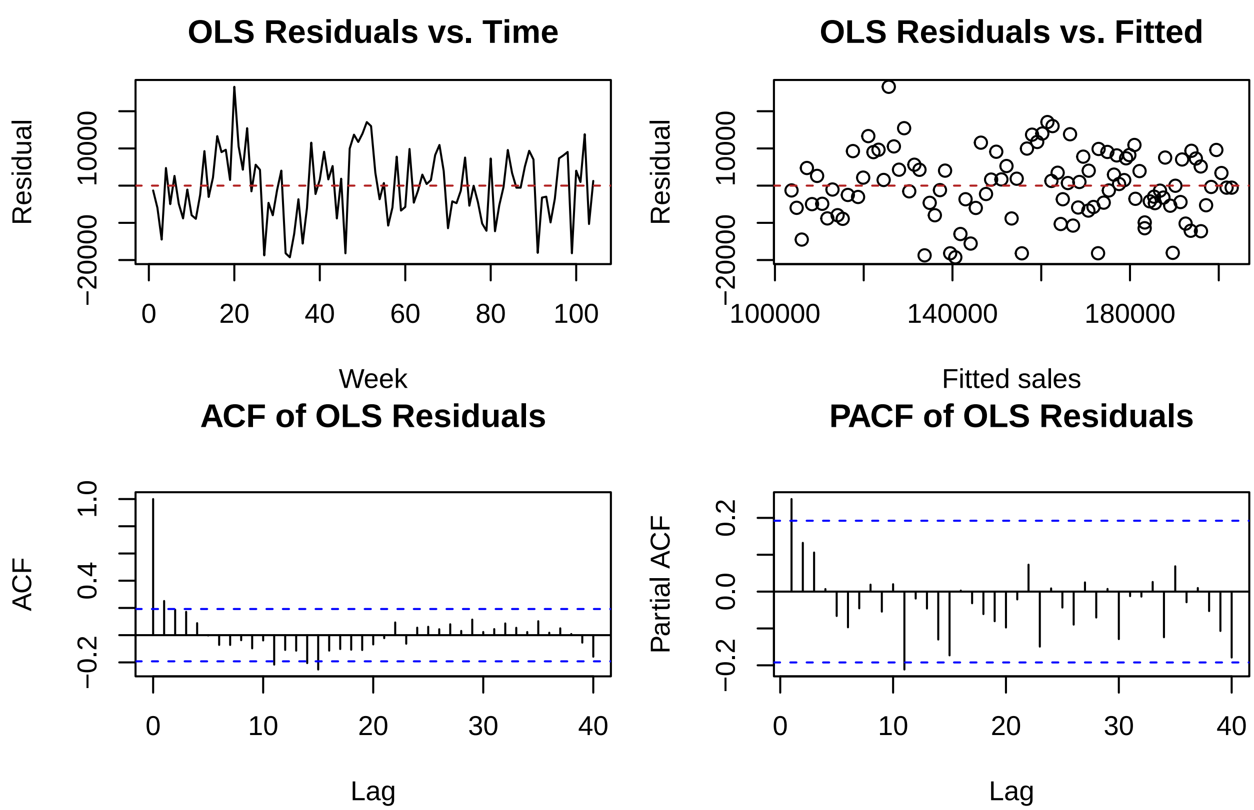
$$y_t = \beta_0 + \beta_1 t + \beta_2 I_t + \beta_3 A_t + e_t.$$

This centered parameterization makes the quantities of interest explicit:

- β_1 is the pre-entry weekly trend.
- β_2 is the immediate level shift at week 88 relative to the pre-entry trend.
- β_3 is the change in weekly trend.
- $\beta_1 + \beta_3$ is the post-entry weekly trend.

An ordinary least-squares fit is useful for inspecting the deterministic structure, but its standard errors are not used for final inference if the residuals are autocorrelated.

Diagnostic	p-value
Ramsey RESET	0.521
Durbin-Watson (positive autocorrelation)	0.002
Nonconstant variance score test	0.291
Shapiro-Wilk	0.514



The Durbin-Watson result and the lag-one autocorrelation show that independence is not a reasonable OLS assumption. The other tests and plots do not flag strong nonlinearity, nonconstant variance, or nonnormality, but those p-values are exploratory because they were computed from dependent residuals. The KPSS test does not reject level stationarity ($p > 0.1$), which supports using a stationary error process without differencing; it does not prove stationarity.

Only two annual cycles are observed. The displayed ACF and PACF cover lags through 40, so annual seasonality at lag 52 cannot be assessed reliably from this sample. No seasonal terms are included.

Joint Regression with ARMA Errors

The regression coefficients and error process are next estimated jointly by maximum likelihood. AR(1), MA(1), and ARMA(1,1) errors are compared because the OLS residual ACF and PACF concentrate most of their structure at short lags.

	Error model	AIC	BIC	Ljung-Box p-value (lag 10)
AR(1)	AR(1)	2196.0	2211.9	0.831
MA(1)	MA(1)	2197.4	2213.3	0.619
ARMA(1,1)	ARMA(1,1)	2195.6	2214.1	0.950

ARMA(1,1) has the smallest AIC, but it improves on AR(1) by less than one AIC point; BIC instead favors the simpler AR(1). The data therefore do not identify a uniquely best low-order error model. ARMA(1,1) is retained as the primary specification because it has the lowest AIC and both error-process coefficients are distinguishable from zero. The AR(1) fit is retained as a sensitivity check.

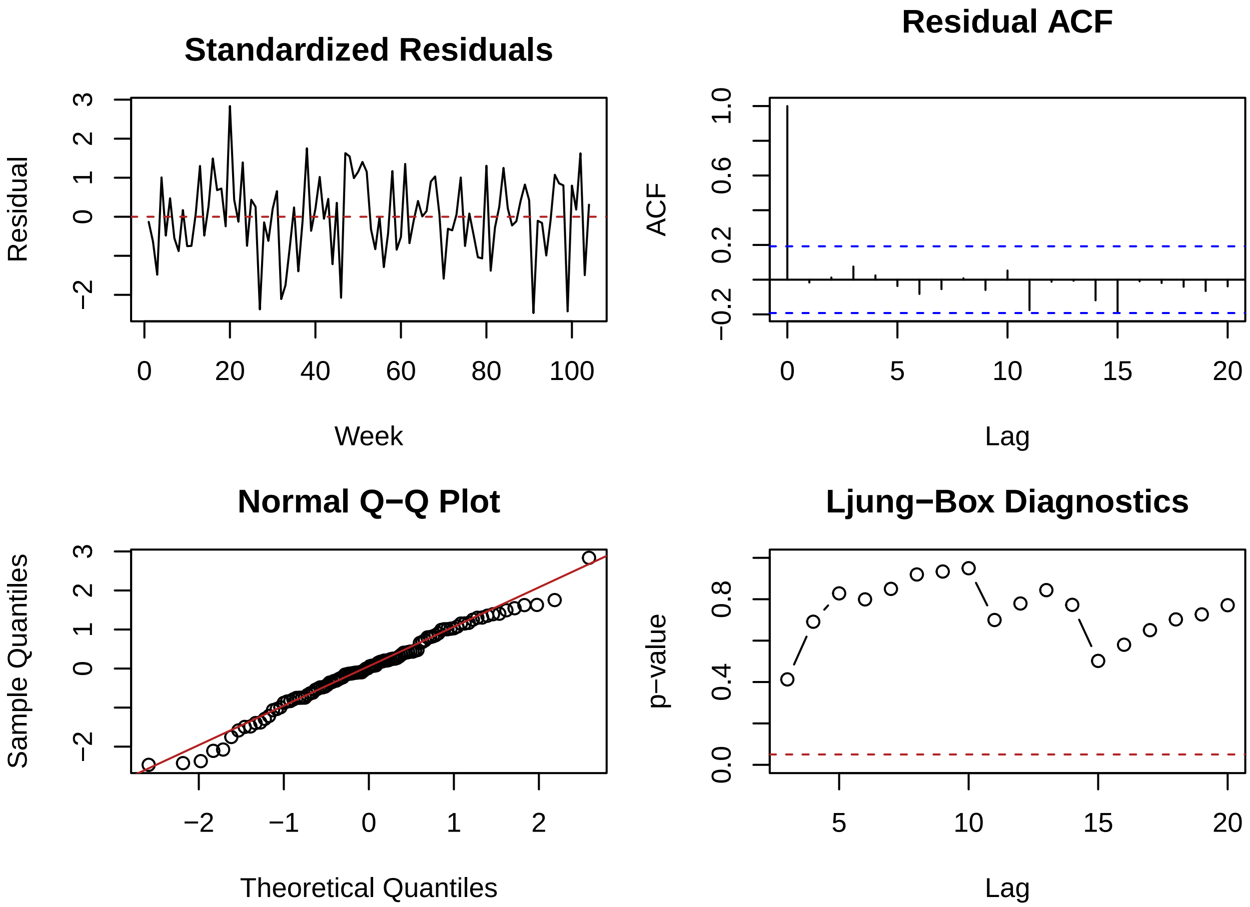
For the selected specification,

$$e_t = \phi e_{t-1} + \varepsilon_t + \theta\varepsilon_{t-1}, \qquad \varepsilon_t \sim \mathrm{WN}(0, \sigma^2).$$

Quantity	Estimate	95% lower	95% upper	p-value
Pre-entry weekly trend	1,153	1,035	1,270	<0.001
Immediate level shift at week 88	-6,545	-18,464	5,375	0.282
Change in weekly trend	-3,390	-4,556	-2,224	<0.001
Post-entry weekly trend	-2,237	-3,387	-1,088	<0.001

The fitted pre-entry trend is approximately 1,153 units per week. The estimated post-entry trend is approximately -2,237 units per week, a change of -3,390 units per week. A likelihood-ratio comparison against an ARMA(1,1) trend-only model supports adding the level and slope changes ($p < 0.001$). The fitted error coefficients are 0.676 for AR(1) ($p < 0.001$) and -0.453 for MA(1) ($p = 0.024$).

The immediate level-shift interval includes zero ($p = 0.282$), so the data do not show a statistically clear jump in week 88. The evidence is concentrated in the change in trend. The AR(1) sensitivity fit produces the same substantive conclusion: an estimated post-entry trend of -2,143 units per week and no statistically clear immediate shift ($p = 0.180$).

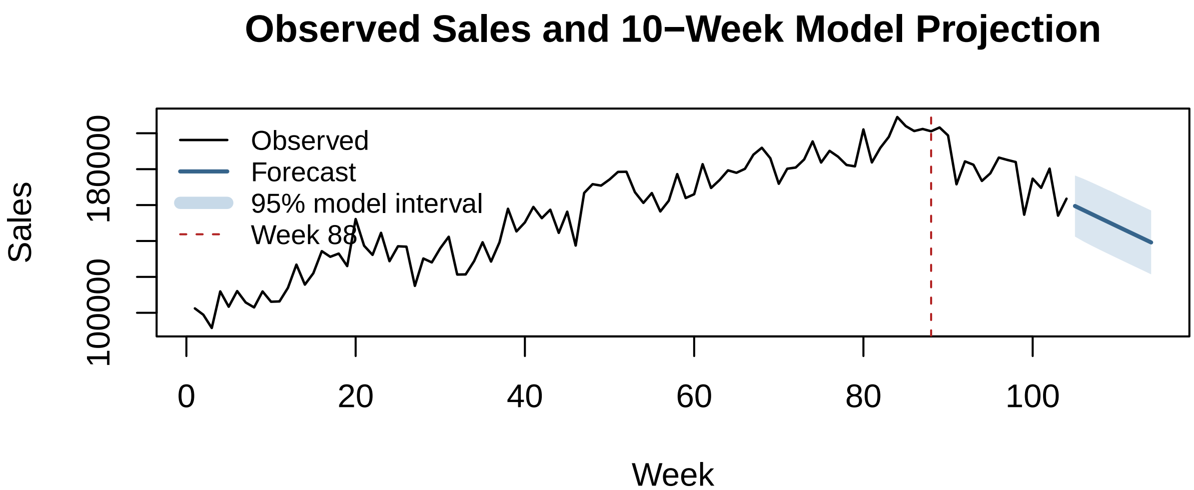


The selected-model residual ACF has no prominent remaining structure, and the lag-10 Ljung-Box test does not reject residual independence (p = 0.950). These diagnostics are consistent with, but do not prove, a white-noise error sequence. The fitted AR and MA roots lie outside the unit circle, satisfying the stationarity and invertibility conditions.

Forecasting

The fitted model is projected ten weeks beyond the observed series. These are conditional, model-based forecasts: they assume the segmented post-entry trend and ARMA error structure continue unchanged.

Week	Forecast	95% lower	95% upper
105	159,465	142,451	176,479
106	157,187	139,755	174,619
107	154,922	137,302	172,542
108	152,666	134,961	170,371
109	150,416	132,673	168,159
110	148,170	130,409	165,931
111	145,927	128,158	163,696
112	143,685	125,913	161,458
113	141,445	123,671	159,220
114	139,206	121,431	156,981



As a limited forecast check, the same ARMA(1,1) specification was refit at ten rolling origins to predict weeks 95 through 104 one step ahead.

Measure	Result
ARMA regression RMSE	13,013
Naive RMSE	15,107
95% interval coverage	8 of 10 forecasts

The model's one-step RMSE is lower than the naive benchmark in this short window, but only 10 forecasts are evaluated and only 87 pre-entry plus 7 post-entry observations are available at the first origin. Coverage is 80%, below the nominal 95%. The ten-week projection should therefore be read as an illustration of the fitted model rather than a validated long-range forecast.

Conclusion

Jointly estimating the segmented regression and ARMA(1,1) errors yields a clear change in weekly trend after week 88. Sales rise by approximately 1,153 units per week before the stated entry date and decline by approximately 2,237 units per week afterward. The immediate level-shift estimate is uncertain and does not distinguish a jump from zero.

The result is robust to replacing ARMA(1,1) errors with the more parsimonious AR(1) alternative. Residual diagnostics do not flag remaining short-lag autocorrelation, while the limited forecast check shows both modest one-step predictive value and substantial uncertainty. Because this is an uncontrolled textbook example with a short post-entry period, the analysis supports an association between the event timing and the brand's sales trend, not a causal claim; market share is not observed.

Data Source

Montgomery, D. C., Jennings, C. L., and Kulahci, M. (2015). *Introduction to Time Series Analysis and Forecasting* (2nd ed.). Wiley. Weekly cereal-sales intervention example and companion data.